

Ciro Villafraz

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AI Engineer

AI & LLM Engineer with 5+ years of experience bridging experimental AI models and robust production environments. Specialized in designing, deploying, and evaluating single and multi-agent systems using Python, LangGraph, and Amazon Bedrock. Proven track record of optimizing LLM inference, building automated safety evaluation systems, and contributing production-quality code to complex, shared architectures. Passionate about applying rigorous agentic workflows and multi-step reasoning to complex real-world and physical constraints.

Work Experience

AI Engineer at Harvard Business Publishing

October 2024 to date

- **Agent Architecture & Deployment:** Designed and deployed scalable agent systems using LangGraph and the Microsoft 365 Agents Toolkit, enabling dynamic, multi-step retrieval and reasoning across 20K+ diverse structured assets.
- **Inference & Decision Optimization:** Optimized prompt templates and query classification networks, drastically improving the agent's autonomous decision-making on tool invocation and parameter selection.
- **Production Reliability & Safety:** Engineered and trained a custom content moderation model (Bagging MLP classifier on 4.6K samples) utilizing pre-trained embeddings to achieve 89% balanced accuracy in edge-case prediction, ensuring safe and reliable agent behavior in production.

Stack: LangGraph, Microsoft 365 Agents Toolkit, Python, FastAPI, Amazon Bedrock, Anthropic, Postgres (RDS), Docker.

AI Engineer at SimSkills

August 2023 to October 2024

A role-playing simulator that uses AI to teach soft skills

- **Multi-Agent Simulation Framework:** Built and operated a robust simulation platform powering 90+ complex agentic role-play scenarios using OpenAI and Anthropic models, achieving an 81% user value rating across 350+ production deployments (including governmental contracts).
- **Experimental to Production:** Spearheaded the transition of experimental agentic RAG and DSPy proof-of-concepts into stable, scalable production workflows.
- **Evaluation & Observability:** Created an automated LLM-based scoring and feedback agent to

evaluate system performance. Implemented LangSmith in production for rigorous tracing, debugging, and continuous improvement of agent reasoning loops.

Stack: Python, LangChain, DSPy, OpenAI, Anthropic, Node.js, GCP, Pinecone.

Machine Learning Engineer at Anyone AI

Jan 2023 to June 2023

- **System Design & API Deployment:** Developed and deployed machine learning APIs using Flask and Docker, conducting rigorous load testing (Locust) to ensure high availability and inference performance.
- **Applied Machine Learning:** Processed a 30+ GB dataset to build a specialized QA pipeline for financial data. Trained and fine-tuned CNNs (ResNet, EfficientNetB0) and supervised models (Random Forest, LightGBM) for classification and risk analysis, achieving high ROC AUC metrics across diverse datasets.

Stack: Python, Scikit-learn, Keras, Haystack, Docker, HuggingFace.

Writing Contributor at Platzi

April 2022 to July 2022

An online courses platform

- Translated complex technical concepts into accessible documentation and articles for 10+ software engineering and data science courses.

Backend Developer at 321 Ignition

Mar 2020 to Aug 2020

A website platform for car dealerships

- **Shared Codebase Contributions:** Developed a centralized GitHub package utilized across multiple backend services, establishing clean architectural boundaries and significantly reducing technical debt for the broader engineering team.
- **Production Testing Pipelines:** Built and maintained comprehensive end-to-end testing pipelines (Cypress) for 10+ dynamically generated production applications, demonstrating a high technical bar for code quality and system reliability.

Stack: Node.js, MongoDB, Docker, GitHub, Cypress.

Fullstack Developer as Freelancer at Talktomira

May 2019 to Jun 2019

- Integrated frontend interfaces with headless CMS architectures (React.js, AWS, WordPress API).

Main Technologies: React.js, AWS, WordPress.

Skills

Agent Tooling & LLMs: LangChain, LangGraph, DSPy, OpenAI, Anthropic, RAG Architectures, VLM Integration, Prompt Engineering, Evaluation Systems.

Software Engineering: Python, JavaScript, SQL, Node.js, REST APIs, System Design.

Infrastructure & Ops: AWS, GCP, Docker, CI/CD, Observability (LangSmith), Vector Databases (Pinecone, Qdrant).

Languages: Spanish (native), English, French, Portuguese.